

Name: _____

FRACTIONS, DECIMALS AND PERCENTAGES

(CONVERTING BETWEEN FDP)

Date: _____

LO: I can convert fluently between fractions, decimals and percentages.

What does converting mean?

Converting means writing a number in a **DIFFERENT** form without changing its value. Fractions, decimals and percentages are three ways to write the same amount.

Key fact: To convert a fraction to a decimal, divide the numerator by the denominator. To convert a decimal to a percentage, multiply by 100.

Equivalent values

●	$\frac{1}{2} = 0.5 = 50\%$	all equivalent
●	$\frac{1}{4} = 0.25 = 25\%$	all equivalent
●	$\frac{3}{5} = 0.6 = 60\%$	all equivalent
●	$\frac{1}{3} = 0.3 = 30\%$	$0.3 \neq \frac{1}{3}$
●	$0.7 = 70\% = \frac{7}{10}$	should be $\frac{7}{10}$

Key idea

Fraction → Decimal: divide top by bottom. Decimal → Percentage: $\times 100$. Percentage → Fraction: write over 100 and simplify.

$$\frac{3}{8}$$

$$3 \div 8 =$$

$$0.375$$

$$37.5\%$$

Divide, then multiply by 100

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 3 - percentage to fraction

Convert 45% to a fraction in simplest form

$$45/100$$

$$\text{HCF of 45 and 100} = 5$$

$$= 9/20$$

$$= 9/20$$

Write over 100, then simplify by dividing both by the HCF (5).

Example 4 - fraction to percentage

Convert $\frac{7}{20}$ to a percentage

$$7 \div 20 = 0.35$$

$$0.35 \times 100$$

$$= 35\%$$

$$= 35\%$$

Convert to decimal first (divide), then multiply by 100.

YOUR TURN – PRACTICE (deliberate practice)

1. Fraction to decimal

Convert each fraction to a decimal by dividing the numerator by the denominator.

- a) $\frac{1}{2}$
- b) $\frac{1}{5}$
- c) $\frac{3}{8}$
- d) $\frac{7}{20}$

2. Decimal to percentage

Convert each decimal to a percentage by multiplying by 100.

- a) 0.4
- b) 0.75
- c) 0.08
- d) 0.125

3. Percentage to fraction

Convert each percentage to a fraction in its simplest form. Write over 100 then simplify.

- a) 20%
- b) 60%
- c) 35%
- d) 84%

4. Fraction to percentage

Convert each fraction to a percentage. Divide first, then multiply by 100.

- a) $\frac{1}{4}$
- b) $\frac{2}{5}$
- c) $\frac{3}{8}$
- d) $\frac{11}{20}$

5. Mixed conversions

Convert each value to the other two forms (fraction, decimal, percentage).

- a) $0.6 = \quad = \quad \%$ b) $72\% = \quad = \quad$ c) $\frac{9}{25} = \quad = \quad \%$ d) $0.875 = \quad = \quad \%$ e) $5\% = \quad = \quad$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{1}{3} = 0.3$ ☐

Correct answer: _____

Reason: _____

b) $0.25 = 25\%$ ☐

Correct answer: _____

Reason: _____

c) $70\% = \frac{7}{10}$ ☐

Correct answer: _____

Reason: _____

d) $0.5\% = \frac{1}{2}$ ☐

Correct answer: _____

Reason: _____

e) $\frac{4}{5} = 0.45$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shop offers a $\frac{1}{4}$ off sale. The sale sign says “20% off everything!”
Is the sale sign correct? Show your working to explain why or why not.

Expression: _____

Simplified: _____

b) Three students scored the following on different tests:
Amy: 17 out of 20 Ben: 0.82 Clara: 78%
Who got the highest score? Convert all to the same form to compare.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step. Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

